Enhancement in LoraWAN's Security With Post-Quantum Key Encapsulation Method

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***Abstract*—Ensuring data privacy and integrity is imperative in wireless connection. Sensitive information needs to be transmitted through long-distance areas in many scenarios as healthcare, agricultural production and smart cities. Not only security, but also performance in this particular type of communication is not a trivial match. The LoRaWAN protocol is capable of transmitting data between IoT as end devices to application servers where the data can be analyzed and stored. The protocol offers a significant security standard and a recognized performance in long-range communication at low cost. However, the advances in Quantum Computing also bring threats to the current state of cryptographic algorithms, possibly compromising communication privacy. LoRaWAN's security is based on AES-128 cryptographic algorithm, which is not safe to quantum computers. Thus, this work proposes an enhancement in LoraWAN's security protocol adding the Kyber KEM-1024 algorithm to reinforce the protocol against quantum-computing attacks. The implementation successfully simulated the communication between an end device and an application server performing the post-quantum algorithm KEM-1024. Despite the results for time of execution can be improved, the proposal presented a safe communication resistant to quantum-computing attacks.**

***Index Terms*—LoRaWAN, post-quantum cryptography, KEM, Kyber, quantum computing**

I. Introduction

Internet of Things (IoT) devices can work as sensors collecting information and transmitting information to a server where applications store and analyze the data to allow more assertive decision-making processes. For that, a long-distance and stable communication system is necessary. Due to the vegetation and terrain density and the long distance itself, not every wireless communication system is adequate. Long Range (LoRa) technology offers great potential and advantages in the agricultural environment [1].

LoRa networks are based on Low Power Wide Area Networks (LPWANs) and offer promising applications for IoT (Internet of Things). Healthcare and, the military are examples other than agricultural areas that leverage Lora's long-range and low-power attributes [2]. The protocol that links LoRa signal to the application is called LoRaWAN (Long Range Network Protocol) ensuring the bidirectional communication between IoT devices [3]. Due to its easy and low-cost deployment and remote maintenance requirements, LoRa technology has been identified as the most suitable LPWAN technology to ensure wireless connectivity for massive IoT networks [4].

IoT devices often transmit sensitive information that must be protected against malicious attacks. However, many threats are still compromising private IoT data, and thus efficient data security schemes need to be adopted [5]. Among the menace to IoT data privacy, Quantum Computing (QC) has been developing at a fast pace with the potential and promise to solve mathematical problems that were considered nearly impossible to solve by a standard computer. This includes the cryptography protocols that are currently in use in the communication in LoRaWAN's protocol.

The National Security Agency (NSA) recognizes and recommends the adaptations of current cryptographic security protocols for quantum-resistant ones known as post-quantum-cryptography (PQC) [6]. Thus, in this work, we propose the security enhancement of LoRaWAN's protocol with a PQC algorithm. The Kyber KEM (Key Encapsulation Method) algorithm is one of the finalists of the National Institute of Standards and Technology (NIST) post-quantum cryptography project and offers significant resistance against quantum-computing attacks. To validate the proposed methodology, we simulated LoraWAN's communication protocol integrated KEM algorithm and measured the time of execution to assess the impact performing in an IoT device.

II. LoRa and LoRaWAN

As the name indicates, LoRa offers a long communication range with a low energy cost. It is a SS-Spread Spectrum modulation and uses a chirp signal varying with the frequency. Because the offset in time and frequency between the sender and receiver is the same, reduces the receiver's complexity and guarantees the low energy cost [3]. As its most important characteristic, the capability of long-distance communication, LoRa can transmit communication through kilometers in a smart city or a farm, for example, [7].

LoRaWAN is the Low Power, Wide Area (LPWA) networking protocol that determines security, network capacity, and quality of service [7], [8]. In general, LoRa is considered as the

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physical layer, while LoRaWAN is the medium access control (MAC) layer of the LoRa stack that adopts a star topology. This configuration is responsible for allowing the communication between multiple end devices (EDs) and the network gateway [4]. LoRaWAN's technology stack can be observed in Figure 1.

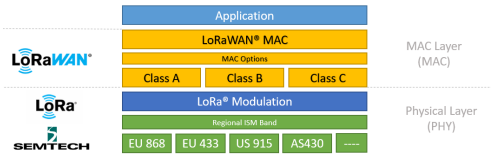


Fig. 1. LoRaWAN Technology Stack [9]

The physical layer (PHY) uses air as a medium for transporting LoRa radio waves from an IoT sender to a gateway receiver and vice-versa [9]. The regional ISM band parameters, defined by EU 868, EU 433, US 915, and, AS430 are the standard channel frequencies that the communication can operate.

The LoRaWAN Media Access Control (MAC) layer, is built on top of the LoRa PHY layer. Here is where the software defines on how devices use Lora's hardware [10]. It is composed of three main components, being the network servers (NS), gateways (GW), and EDs [4]. It also specifies three classes of devices A,B, and C that support bi-directional communication. Class A devices can send an uplink message at any time and open two windows to receive downlink messages from the network. As an extension from the previous, Class B also periodically opens receive windows for downlink messages called ping slots. And, as another extension of Class A, Class C keeps the receive window opened, allowing this class of devices to receive downlink messages at any time [10]

LoRaWAN is established as a star architecture as Figure 2 shows. It allows EDs to send data to multiple GWs, which can send data to a distant NS, where security verification, among other activities can be done [7].

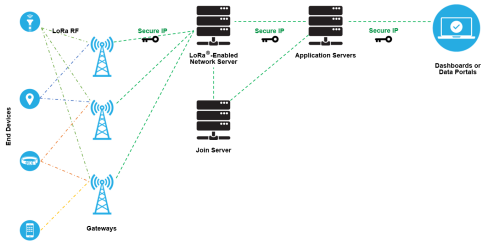


Fig. 2. LoRaWAN Architecture [9]

Surrounding LoRaWAN communication, there is valuable information being traded and thus, security and privacy threats as well. Among the most common privacy violations is the Replay attack. This attack is composed of an ED sending a join-request message to the NS. The attacker intercepts the message and jams the channel so the ED does not establish the communication. After a timeout for communication, another message is sent from the ED. The attacker replays the first join-request message and it is accepted by the NS since it is the first time it is being used.

Another example is the Beacon synchronization attack. An attacker can compromise the GW and try to establish communication with multiple EDs. This will start the protocol in the EDs, but without further communication confirmation, none will be successful. The several communication attempts increase the collisions between the transmitted packages, overcrowding the network [5].

LoRaWAN's cryptography is based on AES-128 and if not safely stored, maintained and tracked, can be broken and reveal information in exploited EDs. As explored by Gunathilake *(et al)* [11], another approach for lightweight cryptography is a necessity for low-power data-processing devices relying in LoRaWAN. One example is the Groover's algorithm, which can reduce the level of security of a cryptographic algorithm, allowing search algorithms to break through security protocols [12]. Therefore, to ensure the communication security in LoraWAN's communication, a PQC algorithm adds a different layer of protection against quantum-computing attacks.

III. Related Works

Lino and Cecílio [13] explore an analysis of multiple cryptography algorithms to be applied in IoT devices within LoRa environment. The work compares the algorithm's performance in 32bit platforms. Code size, execution time, and, efficiency were the criteria evaluated and multiple algorithms. The simulation was implemented using the same chip used in the LoRa network encrypting 100 bytes of data with a 128-bits key. Due to the encryption time required by the algorithms being close to the time necessary to transmit the data, the authors recommend caution when designing IoT systems with the selected algorithms. Despite performing satisfactorily, in this study, the 128-bit key size can be broken by Shor's algorithm. Hence, being vulnerable to quantum computing attacks.

An Asset Tracking System (ATS) based on LoRa using AES-256 cryptography algorithm was implemented by Amelia and Ramadhani [14]. The size of the ciphertext is approximately 64 bytes and it was simulated as an eavesdropping attack. The goal of the simulation is to identify if the ATS system can identify if the communication data was stolen. In the scenario where the AES-256 was tested, it was concluded that the attacker could not access the information due to the encrypted message.Considering that Groover's algorithm mentioned in the previous section, can reduce the security of AES-256 so a search algorithm can break security, the method is not considered resistant against quantum-computing attacks.

A monitoring system based on LoRa using AES-256 and SHA-256 algorithm was implemented by Allamanda *(et al)*

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[15]. In this case, a distributed system of sensors collects real-time data and sends it through LoRa to a central station. The use of SHA-256 and AES-256 is to maintain data integrity against eavesdropping attacks. As the data sent is encoded, at the step when the message is intercepted, the attacker cannot access the data since it is encrypted. However, it was observed that there was an increase in processing time after AES was implemented. Additionally, flash memory and RAM were also evaluated and AES implementation increases mildly its usage.

Pucket (et al) [16] propose hardware accelerators to make it more viable to implement more sophisticated cryptographic algorithms in LoRaWAN. The cryptographic hardware accelerator addition to LoRaWAN protocol proposed by the authors increased the security against known vulnerabilities while reducing energy consumption and processing time. This improvement has a financial trade-off since the accelerator would increase the solution's cost of implementation. This work demonstrates that restrained devices can have improvements in the hardware to comport more expensive cryptographic algorithms and it is worth to enhance the security.

Security enhancement in LoRaWAN by the use of Public Key Cryptography (PKC) was studied by Maarlind *(et al)* [17]. The authors proposed a new protocol using PKC in the activation method. It allows a device to dynamically obtain new root keys and have the ability to replace them as needed. The protocol uses Elliptic Curve Diffie-Hellman (ECDH), while NS uses a static public key and the ED uses an ephemeral public key. The experiment evaluated the battery consumption for transmitting data. It was concluded that the protoclol was feasible and increased the overall LoRaWAN's security. However, this method increased the size of the keys and, consequently, the battery usage as well. Considering advances in IoT devices that have more power at disposal, the trade-off between increasing security and battery consumption can be easier to do in the future with more resources.

One of the main vulnerabilities that threaten LoRaWAN's privacy is the Man in the Middle (MitM) attack [18], specially in the phisical layer. Between the communication of two peer modules, an attacker can intercept, decode, and modify the data. The experimentation was applied appropriately to LoRaWAN hardware and simulated the attack on the communication. The proposed mode uses a 128-bit key cipher and the AES algorithm to encrypt the messages between the two simulated LoRa parties. The implementation showed that even though the data is modified by the attacker, the receiver can identify it when trying to authenticate the communication. Although the method worked to ensure that the receiver is aware of possible modified data being communicated, the GCM, in this case, uses the AES128, with a 128-bit restriction. Even if the cryptographic algorithm allows a higher key size as 256 bits, it is not resistant to quantum-computing attacks.

None of the proposed works present a solution that is resistant to attacks from quantum computers. This fact is worrying the industry due to the exponential increase in LoRa devices on the market and, consequently, the amount of sensitive data that is transmitted. This highlights the importance and urgency to research and develop post-quantum solutions in our communication systems, among them, LoRa.

IV. Post-Quantum Cryptography

Post-Quantum Cryptography (PQC) has the goal to develop system resistant against quantum and classical computers. In a near future, quantum computers will have sufficient resources to break through our modern cryptographic infrastructure and compromise ways of communicating [19]. Therefore, the National Institute of Standards and Technology (NIST) initiated a process to evaluate and standardize post-quantum resistant algorithms for a post-quantum world. In this Session, we describe the public-key finalist CRYSTALS Kyber, the public-key algorithm chosen for the purpose of this work.

Kyber is a Key Encapsulation Mechanism (KEM) cryptographic algorithm finalist of the National Institute of Standards and Technology (NIST) post-quantum cryptography project [20]. Its version, KEM-1024 approximates its security level to AES-256 with the advantage of being resistant to quantum-computing attacks. KEM is composed of probabilistic algorithms represented by := signal and a message *m* formulated as Kyber = (KeyGen, Encr, Decr). This work introduces the Chosen Plaintext Attack (CPA) secure portion of the methodology and after applying a variant of the Fujisaki–Okamoto transform to create the Chosen-Ciphertext Attack (CCA) secure KEM.

Kyber CPA is composed by key generation, encryption, and decryption of the message [21]. While the CCA portion of the method is formed by the key generation, encapsulation and decapsulation of the message.

The key generation algorithm needs some initial parameters k as the message length in bits, q, d_t, d_u, d_v and n. For the purpose of this work, we considered the KEM-1024 parameters described by Avanzi (et al) [22]. The key generation described by Algorithm 1 initiates with the attribution of a random binary string of length of 256 bits to ρ and σ. Next, the matrix A is sampled with the dimensions k x k from the random string ρ. The random string σ is also sampled from the multiplication of the binomial distribution $\beta_\eta^k \times \beta_\eta^k$ as specified by Avanzi. The function Compress receives the sampled matrix *A*,the sampled vectors *s* and *e* together with parameter d_t to output an integer $\{0,...,2^d\text{-}1\}$, where d $< log_2$(q). Finally, the algorithm outputs the private key *pk* the secret key *sk* with a random 256-bit noise vector z .

Algorithm 1 Key Generation

1: ρ , $\sigma \leftarrow \{0,1\}^{256}$
2: $A \sim R^{k \times k}$:= Sam(ρ)
3: *(s,e)* $\sim \beta_\eta^k \times \beta_\eta^k := Sam(\sigma)$
4: $t := Compress_q(As + e, d_t)$
5: return $(pk := (t, \rho), sk := s + z)$

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The Encryption Algorithm 2 starts by attributing a random vector of 256-bit to *r*. The *Decompressq* function is defined by Equation 1 used to calculate t. The next step calculates *A*, which is a *k* by *k* matrix of ρ sampled. The values of r, e_1 and e_2 are derivatives of sampling *r* as step 3 of Algorithm 1. The function Compressq used to compute *u* and υ is described in Equation 2 and, on row 6, is where the initial message *m* is encrypted. These factors are part of the cyphertext *c* that was generated in the encryption phase.

Algorithm 2 Encryption

1: $r \leftarrow \{0,1\}^{256}$
2: $t := Decompressq(t, d_t)$
3: $A \sim R_q^{kxk} := Sam(\rho)$
4: $(r, e_1, e_2) \sim \beta_\eta^k \times \beta_\eta^k := Sam(r)$
5: $u := Compressq(A^T r + e_1, d_u)$
6: $\upsilon := Compressq(t^T r + e_2 = [q/2]m, d_v)$
7: return c:= (u,υ)

$$Decompressq(x, d) = (q/2d)x \tag{1}$$

$$Compressq(x, d) = [(q/2^d)x] mod 2^d \tag{2}$$

Algorithm 3 is responsible for decrypting the cyphertext and starts by applying the Decompressq equation and updating the u and υ values from *c* using the parameters d_u and d_v. The algorithm returns the result from the Compression equation, which is the original message *m*.

Algorithm 3 Decryption

1: u := $Decompressq(u, d_u)$
2: υ := $Decompressq(\upsilon, d_v)$
3: return m:= $Compressq(\upsilon - s^T u, 1)$

The CCA portion of KEM, where an alternate form of Fujisaki–Okamoto transformation is applied to Kyber CPA starts with the key generation Algorithm 1 as well and starts Algorithm 4 for the encapsulation [23]. Initially, as in Algorithm 2, a random *m* vector of 256 bits is generated. Followed by a hash function *G* using the previous calculated *pk* and *m* generating the pair *(κ,r)*. Algorithm 2 is applied to the public key composed by the pair *(t,ρ)*, *m* and *r*, which is derived to *c*. A hash function is applied to *c* and κ to achieve *K*. The algorithm outputs the pair *c* and *K*.

The Decapsulation Algorithm 5 starts by applying Algorithm 3 to the secret key *sk* and the pair (u,υ) to generate *m'*. The following pair (κ', r') is calculated as in Algorithm 4. And, (u',υ') is computed by applying Algorithm 2 to the public key *(t,ρ)*, *m'* and *r'*. A verification now takes place where if $(u', \upsilon') = (u, \upsilon)$, then the decapsulation is successful.

Algorithm 4 Encapsulation

1: $m \leftarrow \{0,1\}^{256}$
2: $(\kappa, r) := G(H(pk), m)$
3: $(u, \upsilon) := Kyber.CPA.Encryption((t, \rho), m; r)$
4: $c := (u, \upsilon)$
5: $K := H(\kappa, H(c))$
6: $return(c, K)$

Algorithm 5 Decapsulation

1: $m' := Kyber.CPA.Decryption(sk, (u, \upsilon))$
2: $(\kappa', r') := G(H(pk), m')$
3: $(u', \upsilon') := Kyber.CPA.Encryption((t, \rho), m'; r')$
4: if $(u', \upsilon') = (u, \upsilon)$ then
5: return $K := H(\kappa', H(c))$
6: else
7: return $K := H(z, H(c))$

V. Proposal

A. LoRaWAN's Cryptography

LoRaWAN's cryptography protocol is based on two root keys *NwkSKey* and *AppSKey* [24]. When a device joins LoraWAN's network, the keys are generated and they are unique per device and session.

As Figure 3 shows, the Network Session Key *NwkSKey* secures the exchange between ED and the network and application service. The *NwkSKey* is a 128-bit key that ensures the integrity of the message using the AES Cipher-based Message Authentication Code (AES-CMAC). It is impossible to temper the encrypted message without *NwkSKey* because the Message Integrity Code (MIC) check will fail [25].

The Application Session Key *AppSKey* is responsible for the transferred data encryption. The data to be transferred or payload is encrypted with the 128-bit *AppSKey* using AES Counter Mode (AES-CTR) [26]. In the case of a message interception, it remains encrypted without the *AppSKey*

As Bonnetain (et al) [27] presented in their work, AES-128 shows to be vulnerable against Groover's algorithm. And, despite increasing the key size, Groover's algorithm performed in a quantum computer can reduce its security level [28]. Additionally, increasing key sizes can affect the performance trade-off in the IoT environment and has to be considered. As concluded by Thaenkaew (et al) [29], escalating AES' key size can occur in a 32% larger encryption time. Besides that, KEM Kyber is one of NIST's choices of post-quantum algorithms to be included in their standards. Thus, its applicability is worth being validated in a LoRaWAN case.

B. Quantum-Resistant LoRaWAN

This work proposes an enhancement in LoRaWAN's cryptographic protocol by adding a PQC algorithm in the communication between the ED and the Application Server. This

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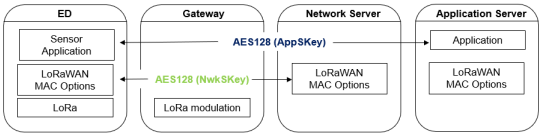


Fig. 3. LoRaWAN Cryptography

implementation simulates a generic and scalable use case where an IoT device, as a sensor, sends data to a server through LoRaWAN network. The implementation of CPA-secure KEM-1024 instead of the AES-128 as the *AppSKey*, enhances the security from 128 to 1024 bits and, adds resistance against quantum-computing attacks as Groover's algorithm. As Figure 4 shows, KEM-1024 is now ensuring the content of the message integrity between the EDs and the Application Server.

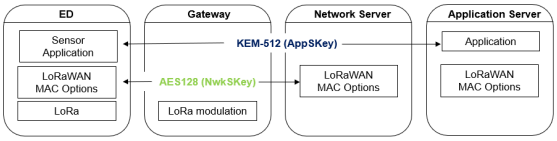


Fig. 4. Quantum-Resistant LoRaWAN Cryptography

To assess the enhancement in LoRaWAN's communication security protocol, a simulation in *python* was implemented as a socket network. The socket network considers a client and a server, being the ED and the Application Server respectively in this simulation. Besides the communication itself, the cryptographic algorithm Kyber KEM-1024 was relied on to ensure the integrity of the transferred message. For that, the open-source *python* library wrapper *liboqs* by open-quantum-safe project (OQS) was implemented in 4 CPU cores and 16 GB memory device [30].

The Kyber KEM-1024 methodology follows as described in Section IV, where the ED generates the public and secret keys with Algorithm 1 and encapsulates the message m using Algorithm 4. This is followed by sending the secret and the encryption pair (u, v) to the Application Server, where the message is decapsulated by Algorithm 5 as represented in Figure 5. This setup can be widely applied in different use cases and in a scalable way allowing many EDs to connect to the Application Server.

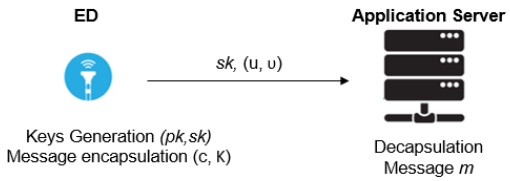


Fig. 5. Kyber KEM-1024

VI. Results

The Quantum-Resistant LoRaWAN's enhanced security protocol was implemented by replacing the AES-128 algorithm as *AppSKey* in the communication between the EDs and the Application Server for the quantum-resistant Kyber KEM-1024. The length of the cyphertext resulted in 1568-bits, the public key in 1568-bits, the secret-key in 3168-bits and, the shared secret in 32-bits. The simulation tested the communication in a socket environment 100 times and the average of the execution time results for each KEM's phase can be observed in Table I.

TABLE I
Simulation Results

Phase	Time of Execution (s)
Key Generation	0.002875
Encapsulation	0.000210
Decapsulation	0.000090

As already mentioned in Session V, simply increasing the key size in AES-128 does not guarantee a quantum-computing attack resistance or a satisfactory performance. Therefore, developing implementations of PQC as KEM-1024 in protocols as LoRaWAN are an important step in the process of ensuring a post-quantum safe communication.

Considering the benchmarks from the work published by Garrach (et al), [31] where the order of the time of execution is in μs, the results obtained still need improvements. One of the reasons for this difference is that in this work, we made use of the *python* wrapper version from OQS project. Since the wrapper needs to call the function native in C, it can cause overhead and compromise its performance for the time of execution.

Nevertheless, depending on the application, the time of execution for Kyber KEM in a fraction of a second can be considered adequate in LoRaWAN's communication. In use cases such as agricultural communication between sensors and a base for example, the time sensitivity does not require μs precision. Enhancing the security of the protocol to ensure a quantum-computing attack-safe message requires a trade-off between the time performance and KEM's implementation. Thus, even though, the performance can be improved, for specific use cases, the proposed enhancement already presents realistic results for an IoT communication scenario through LoRaWAN.

VII. Final Considerations

This work proposed an enhancement in the LoRaWAN cryptographic protocol replacing the *AppSKey* from AES-128 to the PQC algorithm Kyber KEM-1024. The goal of the work is to implement successfully a PQC algorithm in the protocol to improve the security against quantum-computing malicious attacks and ensure the message integrity.

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The algorithm implementation occurred in a *python* socket application simulating the communication between an ED and the Application Server. This scenario can be translated and replicated in many use cases as agricultural communication, healthcare, smart cities and many others. The Kyber KEM-1024 implementation was successful and the message privacy was guaranteed with significant improvement considering the key size. The algorithm's time of execution was measured during the simulation. The time of execution was considered appropriate for sensor communication in a non time-critical scenario for a resource restrained IoT device.

Considering benchmarks for the time of execution found in the bibliography [31], [32], the results obtained in the simulation could be more satisfactory. As a future work, the algorithm performance can be improved to expand the range of use cases. Besides that, with a reasonable improvement in the performance, replacing the *NwkSKey* by Kyber KEM or a digital signature algorithm can enhance the security between ED and Network Server as well.

Despite that, the enhancement had been proven to work in a LoRaWAN communication simulation providing an additional layer of security not only in the size of keys but also against quantum-computing attacks. The topic comes with various challenges and trade-offs. Nevertheless, ensuring quantum-computing attack resistance is imperative from now on. The advances in quantum computing shed a light on the urgency of implementing quantum-resistant algorithms in our communication networks.

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